

Tuning Matters: Classical Models on Imbalanced and High-Dimensional Data

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Fictional paper written to test Preflight. Not a real publication.

Abstract

Classical models remain strong baselines on tabular data, provided they are tuned. We study logistic regression, a tuned RBF support vector machine and nearest-neighbour classification on four synthetic benchmarks that mimic common failure modes of real data: class imbalance, many features with few samples, and very large sample sizes.

1 Experimental protocol

Every number in this paper is the mean of 10 runs with random seeds 0 to 9. Each run draws a stratified 75/25 train/test split. Features are standardised where stated.

2 Experiments

SynthMed is an imbalanced diagnosis-style task with 15% positives, so we report macro-averaged F1. Logistic regression is regularised with $C = 1.0$ and trained for at most 1000 iterations on standardised features.

SynthHD has 500 samples and 150 features. The SVM is tuned over C and gamma by 3-fold cross-validation. The logistic regression baseline uses $C = 1.0$ and up to 2000 iterations.

On SynthKNN-Aug we use a 1-nearest-neighbour classifier on standardised features. The data pool is enlarged by re-sampling 40% of the samples.

SynthLarge-2M has 2,000,000 samples with 40 features; logistic regression uses $C = 1.0$ and at most 1000 iterations.

3 Results

Table 1. Macro-F1 (%) on SynthMed.

Method	SynthMed macro-F1
Logistic regression	90.97

Table 2. Test accuracy (%) on SynthHD.

Method	SynthHD accuracy
SVM (RBF, tuned)	86.64
Logistic regression	85.12

Table 3. Test accuracy (%) on SynthKNN-Aug and SynthLarge-2M.

Method	SynthKNN-Aug	SynthLarge-2M
1-NN	90.60	
Logistic regression		81.08

The tuned SVM consistently outperforms logistic regression on SynthHD, by 1.52 points. Nearest neighbours profit from the enlarged data pool, and logistic regression scales to two million samples

without loss of accuracy.